Computer Communications 154 (2020) 278–287

Contents lists available at ScienceDirect

Computer Communications

journal homepage: www.elsevier.com/locate/comcom

Mining of instant messaging data in the Internet of Things based on support vector machine

Yang Chen

Network Data Centre, Communication University of Zhejiang, HangZhou 310014, China

ARTICLE INFO

Keywords:
Support vector machine
Internet of Things
Big data
Information classification

ABSTRACT

With the development and popularization of mobile communication technology, mobile communication has been applied in more and more industries, and has effectively promoted the progress of the times. How to take effective methods and means to ensure the correct classification and mining of effective instant messaging information has become a top priority for major operators. Based on the analysis of the theory and feature selection technology of support vector machine in machine learning field, this paper studies the main parameters that affect the feature selection of support vector machine, aiming at the dimension disaster caused by large-scale data volume and redundant features of the Internet of things. In this paper, an algorithm is proposed to optimize the feature subsets of samples, and then add the parameters of support vector machine to optimize the classification. The experimental results show that the algorithm has a good effect on the classification of effective instant messaging information of Internet of things big data, and has a good effect and practical application value.

1. Introduction

In recent years, computers are gradually popularized in people's life, which not only promotes the improvement of communication network technology level, but also promotes the rapid development of international Internet [1–4]. Internet has become the main way of global information dissemination and the largest information resource base in the world. Internet applications cover all aspects of life, such as instant messaging, email, social networks, news sites, online forums and so on, as shown in Fig. 1. We are used to relying on the Internet to provide convenient information to understand the world and maintain interpersonal communication [5,6]. Most of the information generated in the process of network interaction is in the form of communication information. In the face of massive communication information, it is just as difficult to get the data we want to know from it as looking for a needle in a haystack [7]. Therefore, how to quickly and accurately find the valuable target category information from a large number of unstructured, noisy, high-dimensional sparse, redundant communication information is the problem we need to solve at present. It is not hard to imagine that if we use the traditional artificial method to search, it will not only cost a lot of time and energy, but also cannot completely traverse all the information contained in the network, so we lead to the communication information mining theory [8–10]. In the era of big data, it is necessary to establish user centered self-service big data governance. Information sorting, data management and control, connecting users, and intelligence are the four main stages of self-service big data governance. Mastering a series of key technologies and technical principles is an important foundation for big data governance [11]. Big data mining of communication information is a process of automatically finding the information with potential relationship through the analysis and operation of a large number of data. It usually combines machine learning, statistical analysis, pattern recognition and other methods to complete the function of data mining. In view of communication information as an important part of network data type, the research on big data mining technology of communication information has become a hot issue in the field of data mining and information search.

Feature selection is actually a kind of data analysis method. The main task is to find out the key features of something different from other things or find out the relationship between features, so as to lay a good foundation for further scientific analysis and decision-making. In the field of machine learning, many researchers take various classifier learning methods as their research topics [12,13]. However, in general, they need to use feature selection to preprocess the data. At the same time, the emergence of high-dimensional data and large data sets has played a role in promoting the research of feature selection. Because feature selection can reduce dimensions, it has become a research hotspot in machine learning field. This paper presents a feature subset optimization method based on support vector machine and genetic algorithm. As a new technology in the field of data mining, SVM has been successfully applied in many fields, such as face recognition, handwritten Chinese character recognition and so on [14,15]. However, the choice of parameters and kernel function parameters in SVM is always

E-mail address: 20030061@cuz.edu.cn.

https://doi.org/10.1016/j.comcom.2020.02.080
Received 9 January 2020; Received in revised form 11 February 2020; Accepted 27 February 2020
Available online 2 March 2020
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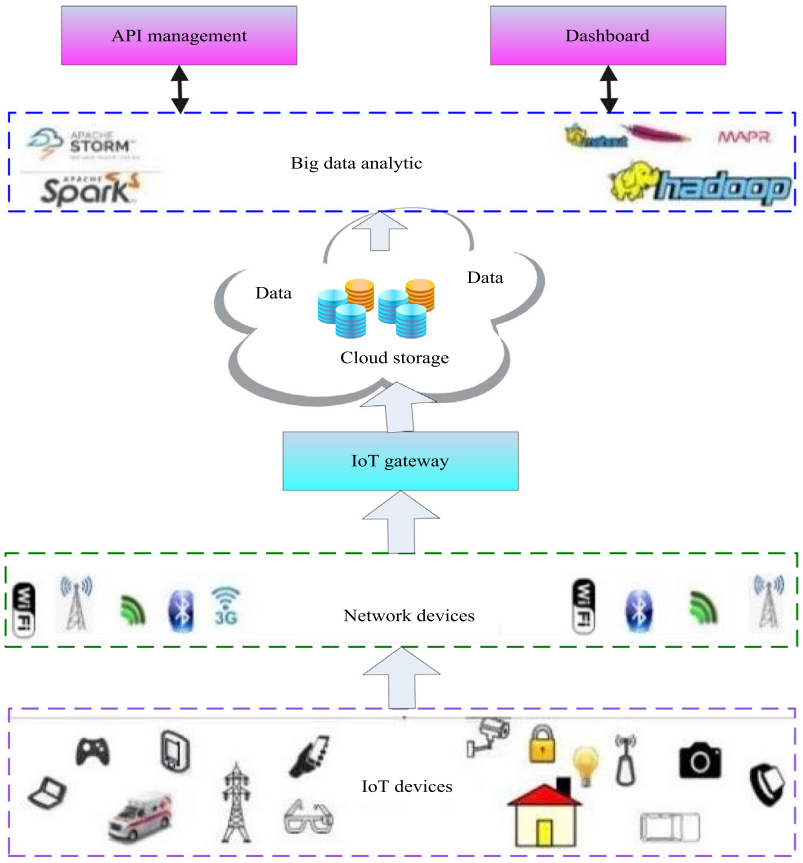


Fig. 1. The working structure of big data under the internet of things.

the main factor that affects the performance of SVM. A reasonable parameter value can make SVM have higher classification accuracy and generalization ability [16]. The traditional method of parameter selection is through repeated experiments, the relatively better parameter value is selected manually, but this method needs higher time cost. In this paper, a new method of SVM parameter selection is proposed. By mixing the parameters of SVM into the big data of the Internet of things, the better parameters and feature subsets of SVM are selected at the same time. We hope that support vector machine, as a method with perfect theoretical basis and easy application, can make a better contribution to the changing practical problems.

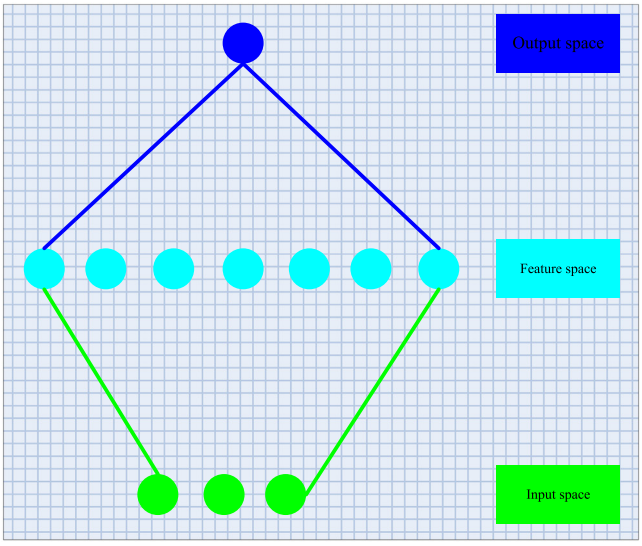
Mobile communication has become one of the industries with the fastest development speed, the fastest technology update and the largest market capacity [17]. As an important business of telecommunication industry, mobile communication directly provides services to the society. Its development trend, scale adjustment and technology upgrading have a crucial impact on national information security, economic security and development, and telecommunication operation [18,19]. And with the popularization of 4G network, all kinds of new services are listed one after another, and the business volume is also increasing. It will become an urgent problem for major operators to ensure the stability of data business and business volume throughout the whole network and extract effective information from massive data. In this paper, a new big data algorithm is used to extract features and predict a large number of core performance indicators in the real network, and support vector machine (SVM) technology and association rule technology are used to further optimize the prediction algorithm. From the network side or user face, we can truly quantify the network and user behavior, mine effective instant messaging information and classify predictive information, so as to ensure the establishment of an efficient, safe and stable mobile communication network.

2. Support vector machine and training algorithm

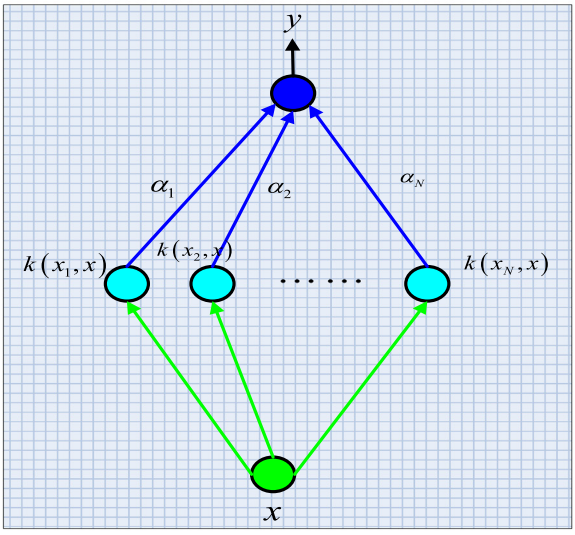
2.1. Support vector machine

Support vector machine belongs to the category of computational learning in artificial intelligence. It is a machine learning technology developed in the mid-1990s. Compared with the traditional learning technology, it has a solid theoretical basis. Seen from a large number of applications, its performance is indeed superior to the traditional methods. It creates and maintains the best record on many specific problems such as handwritten numeral recognition, text classification, etc. Support vector machine has strong learning ability and generalization ability. Its application prospect is very good [20]. So far, it has involved many fields such as pattern recognition, regression estimation, data mining, information retrieval, intelligent signal processing, intelligent control, nonlinear pattern reconstruction, etc.

The research on this theory began in the late 1960s. In the past 30 years, the former Soviet scholar Vapnik has done a lot of groundbreaking work and creatively put forward the principle of “structural risk minimization”. The work in this period is mainly theoretical. According to this principle, it can not only fit the known training data well, but also fit the unknown or non-existing data well, both of which are considered, so it has a strong generalization ability [21,22]. This is related to the structure of learning system. With the successful application of SVM more and more people join in the research, forming the research hotspot in the field of machine learning in recent years. Basic research involves training algorithm, model selection, multi-target classification, and application research involves three-dimensional target recognition, speech recognition, nonlinear pattern reconstruction, intelligent signal processing, information retrieval, biological information processing, text classification, data mining, etc. Let F represent the input vector, f is the feature space (high dimensional inner product



(a) Schematic diagram of support vector machine



(b) Structure chart of support vector machine

Fig. 2. Schematic diagram of support vector machine.

space), the feature vector $Z_k \in F, Z_k = \Phi(x_k)$, and Φ is the feature map. Kernel function $K(x_k, x_l) = \Phi(x_k) \cdot \Phi(x_l)$, "·" represents inner product in F. The optimization problem of $\{z_k\}$ linear separable time and support vector machine is:

$$\min \frac{1}{2} \|w\|^2 \tag{1}$$

When $\{z_k\}$ is non-linear and separable, the second penalty is used, and the related optimization problem is:

$$\min \frac{1}{2} \|w\|^2 + \frac{c}{2} \sum_k \xi_k^2 \tag{2}$$

SVM is characterized by training samples and kernel functions. The schematic diagram of support vector machine is shown in Fig. 2.

The functions of SVM include: Classification (recognition), prediction, clustering, anomaly detection (novelty detection), density estimation, etc.

How SVM works:

(1) Collect training samples and select kernel function.

(2) Using training algorithm, support vector machine is constructed automatically.

(3) Input the unknown (New) data into SVM, identify and predict according to the output.

2.2. Support vector machine training algorithm

The support vector machine method which is suitable for solving two or more kinds of problems is worth our in-depth study. However, according to the research content of this paper, it can be applied to the research based on two kinds of problems. Other problems in support vector machine, such as model selection, are also related to the training algorithm. Therefore, the study of SVM training algorithm has both theoretical significance and application value. The training of support vector machine comes down to solving quadratic programming problem, but the Hessian matrix of the problem is usually dense, so the storage cost is very high when dealing with large-scale problems [23]. For example, when the number of samples reaches 4000, 128M memory is not enough, so the classical optimization method is not applicable. It is the goal of people to develop the algorithm with short time and less memory.

Through in-depth analysis, it is found that the design of NPA algorithm is unreasonable, which affects the speed of the algorithm. In view of these unreasonable points, the corresponding improvement measures are proposed, and the improved training speed is significantly higher than that of NPA. A relatively low computational complexity and relatively short time-consuming support vector machine algorithm is particularly important for solving big data problems. Trying to find a fast and effective classifier suitable for multi value classification will become the focus of future research.

Each cycle of NPA algorithm consists of two test steps and one iteration step. First, the first type of test is carried out to find the index k that does not meet the termination condition in the non-support vector, and then the second type of test is to find the index k that does not meet the termination condition in the support vector [24,25]. If such K is found in a certain test process, it will enter the iteration step, and then enter the next cycle. This continues until the indicator k that does not meet the termination condition cannot be found in a cycle. In order to reduce the calculation cost, only one test of the first type is carried out in each cycle, while the second type of test is carried out several times continuously until there is no K in the support vector that does not meet the termination condition, or the number of tests reaches a preset threshold, that is to say, another sub cycle is formed on the support vector. This is very important to improve the overall performance of the algorithm.

The iterative steps of NPA algorithm are divided into the following four situations, as shown in Fig. 3, which are processed separately.

A lot of experiments have been done with NPA algorithm. The results show that with the increasing of penalty coefficient, the cumulative times of the second kind of test and its accompanying iterations increase dozens of times and hundreds of times, while the cumulative times of the first kind of test and its accompanying iterations are relatively stable [26,27]. This shows that the first kind of test focuses on the initial stage of the whole learning process, and the rest of the time is spent on the second kind of test and its accompanying iteration. In this case, it is not the best policy to share the same iterative process under two kinds of tests. If the iterative process under the second kind of test is further optimized, the performance of NPA will be improved.

In the second kind of tests, although more detailed calculations are carried out and more information is obtained, these information are not fully utilized in the subsequent iterations.

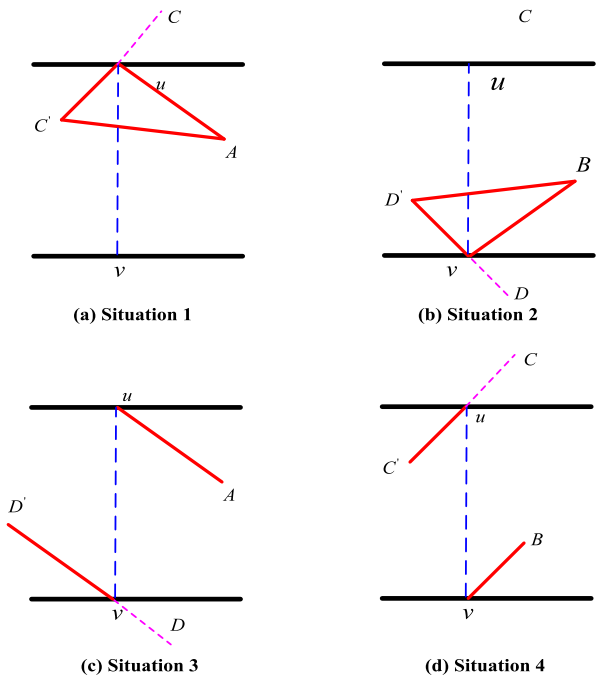


Fig. 3. Four cases of iteration.

3. Communication information selection method for big data text feature

In this paper, the traditional algorithm of each module of instant messaging information classification system is studied and analyzed, which is improved in combination with practical application to make it more suitable for information mining and classification of big data. The overall architecture of the classification system designed in this paper is shown in Fig. 4. In the future work and learning, we need to carry out many times of simulation and verification, and finally establish a relatively optimized network information extraction system.

3.1. Big data feature selection method

Feature selection is a process of selecting the most representative feature subsets from the initial complete feature set, which is a key step in the process of text classification. By reducing the dimension of feature space, not only the calculation process is simplified and the classification speed is improved, but also the classification accuracy is improved by adjusting the feature set. The feature selection method analyzes the features and designs the calculation method of the relevance degree of the specified category documents, calculates and sorts all the features in the thesaurus, and selects the feature items by setting the threshold value. Common feature selection methods mainly include the following:

Document frequency refers to the proportion of the number of documents containing characteristic ti in the sample set. The formula is as follows.

$$\mathrm{DF}\left(t_i\right) = \frac{df\left(t_i\right)}{N} \tag{3}$$

In the formula, $df\left(t_i\right)$ represents the number of documents containing characteristic it, and N represents the total number of documents in corpus. The specific method is to calculate the document frequency corresponding to all the features of the corpus, that is, the entries whose document frequency is lower than the threshold value are filtered according to the preset threshold value, and some entries whose document frequency is higher are retained. The selection of threshold should consider two aspects: one is to be able to remove the feature words (low-frequency words) in a small number of documents in corpus; the other is to consider that the words in most documents do not contribute much to the classification of documents, and should also be deleted. The method based on document frequency is simple and low complexity, which is suitable for large-scale data sets. Its disadvantage is that it may delete some feature words with low frequency but containing important category information, which will affect the classification effect.

Mutual information is also used to measure the degree of association between a feature t and a designated category c. If the amount of mutual information is larger, it means that the more likely the feature t is to appear in this category, the more representative it is. The specific calculation formula is as follows.

$$\mathrm{MI}\left(t, c\right) = \log_2 p\left(t|c\right) - \log_2 p\left(t\right) = \log_2 \frac{p\left(t|c\right)}{p\left(t\right)} \tag{4}$$

The disadvantage of mutual information is that it ignores the frequency of features appearing in each document, so it will lead to a bias of low-frequency words in the method based on mutual information. If only relying on this method to select feature subset, it will filter out many high-frequency valuable features, and affect the classification effect.

After the implementation of text classification algorithm, we need to evaluate the classification effect, and use the authoritative evaluation indicators or criteria as guidance. The evaluation process is to test the judgment result of the input test text category by the classifier. The commonly used evaluation parameters include recall rate, precision rate value, etc. in addition, for multi classification, there are three evaluation parameters for micro average and macro average. This paper introduces the common evaluation criteria to verify the performance of the classification system.

The big data algorithm can detect the outliers in the initial data, get the result set of outliers, and process the error data for all outliers. The processing methods of abnormal data generally include mean value method, maximum likelihood estimation method, multiple interpolation method, etc. the error data method used in this paper is to construct linear fitting curve, which has the advantages of simple calculation, small error and stable performance. The specific method is to take the outliers as the center point, take the first several outliers of matrix B as a data sequence, and make a linear regression of least square method. The substitution value of outliers is calculated by linear regression equation, and the new matrix is obtained by substituting the original outliers of matrix.

3.2. Diversity design method of individual classifier

The diversity among classifiers has a great influence on the integrated classifiers. How to design a diversity classifier? At present, the design methods of diversity member classifiers are mainly as follows:

Different sets of training samples are used to train the classifier. Through the sampling of training samples, different data subsets are formed. Different data subsets are used to train the same type of classifiers or different types of classifiers, so as to realize the diversity of classifiers.

The same training samples are used to train different types of member classifiers. At present, there are many classification methods, such as decision tree, k nearest neighbor, neural network, Bayesian classification, support vector machine, etc. different classification methods have different training results for the same training set and the output representation is the same. Therefore, the results of classification should be combined in a unified way.

According to the characteristics of sample attributes, the training set is divided into different data subsets, and then the classification method is used to train the member classifier. Firstly, the large feature space is divided into several small feature spaces, and then the member classifiers are constructed on each small feature space, and then these

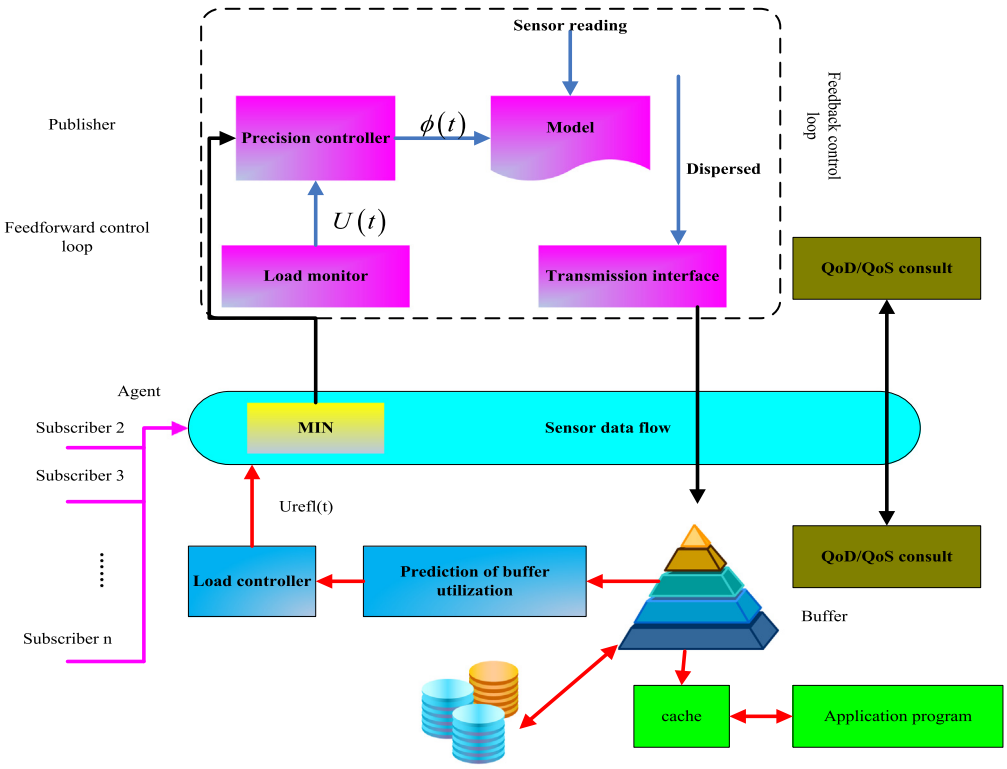


Fig. 4. The model of big data information feature control system in the environment of Internet of things.

member classifiers are integrated, that is, the diversity of member classifiers is realized by splitting sample attributes.

Consolidation category label. When the total number of training samples is large, the training samples are first divided into two subsets according to the class label, and the training samples in the same subset are regarded as a class. Then, a member classifier is generated by using the classification method, which is repeated and randomly combined many times to generate multiple member classifiers.

Change the training parameters or the structure of the classifier. For example, in SVM, different member classifiers can be established by changing kernel function or kernel function parameters.

For the multiple member classifiers obtained by training, it is not that the integrated classifier obtained by combining them all is the best, so it is necessary to select the member classifiers. For a small number of classifiers, a comprehensive search method can be used to select the best combination from all the integrated classifiers. However, when the number of classifiers is large, the complexity of search limits the scope of application of comprehensive search, so there should be certain criteria to select the member classifiers for integration. The diversity of member classifiers is an important selection criterion, which is the best choice to ensure the generalization ability. At present, there are many methods to measure diversity. Kuncheava et al. Classified diversity metrics into “paired metrics” and “non-paired metrics”. The pairwise metric is used to measure the classification difference between any two classifiers in the integrated system, while the non-pairwise metric is used to calculate the group diversity as a whole.

(1) Inconsistent measures

This method uses the ratio of samples with inconsistent output of classifier and mm to all samples as the diversity measurement standard of these two classifiers.

$$S_{i,j} = \frac{N^{01} + N^{10}}{N^{11} + N^{00} + N^{01} + N^{10}} \tag{5}$$

The value range of $S_{i,j}$ is from 0 to 1. When the value of $S_{i,j}$ is 0, the prediction results of classifier M_i, M_j are the same, that is, the classification is correct or wrong at the same time. When the value of $S_{i,j}$ is 1, the prediction result of M_i, M_j is different.

The integrated inconsistency measure is also the average among member classifiers, that is:

$$S = \frac{2}{M(M-1)} \sum_{i \neq j} S_{i,j} \tag{6}$$

(2) Correlation coefficient method

Correlation coefficient method is often used in continuous variables, which is a widely used statistical method.

$$\rho_{i,j} = \frac{N^{11}N^{00} - N^{01}N^{10}}{\sqrt{\left(N^{11} + N^{10}\right)\left(N^{01} + N^{00}\right)\left(N^{11} + N^{01}\right)\left(N^{10} + N^{00}\right)}} \tag{7}$$

The integrated correlation coefficient method is also the average value between member classifiers, that is:

$$\rho = \frac{2}{M(M-1)} \sum_{i \neq j} \rho_{i,j} \tag{8}$$

(3) Double error metric

This method focuses on the samples which are misclassified by M_i, M_j.

$$W_{i,j} = \frac{N^{00}}{N^{11} + N^{00} + N^{01} + N^{10}} \tag{9}$$

The formula for calculating the integrated diversity value using this measurement method is:

$$W = \frac{2}{M(M-1)} \sum_{i \neq j} W_{i,j} \tag{10}$$

(4) Kappa metric

The measurement formula of this method is as follows:

$$Ka_{i,j} = \frac{2\left(N^{11}N^{01} - N^{10}N^{00}\right)}{\left(N^{11} + N^{10}\right)\left(N^{00} + N^{01}\right) + \left(N^{11} + N^{01}\right)\left(N^{10} + N^{00}\right)} \tag{11}$$

The formula for calculating the integrated diversity value Ka is as follows:

$$\mathrm{Ka} = \frac{2}{M(M-1)} \sum_{i \neq j} Ka_{i,j} \tag{12}$$

3.3. Training classifier module

The training module is the largest part of the whole classification process, which determines the complexity of the algorithm. The quantity and quality of training samples have a great influence on the performance of classification function. The more training samples, the richer the content of positive and negative samples, the higher the accuracy of classification function. In order to improve the accuracy of classification algorithm, we add the process of parameter optimization and cross validation in SVM training.

The common methods of parameter optimization in SVM include experiment method, mesh search method, genetic algorithm, particle swarm optimization method, etc. The experimental method needs a lot of time to find the optimal solution. GA and PSO are more complex to realize, and they are easy to be limited to the local optimal solution. The grid search method is to divide the value range of the parameter group to be optimized into grids within a certain range, and obtain the optimal parameter array by traversing each point of the grid. The grid search method is easy to understand and operate, and is suitable for solving the optimization problem in the case of small samples. Therefore, this paper chooses this method to optimize the parameters of SVM kernel function.

In this paper, the radial basis function is selected, which has better classification performance and learning ability than other kernel functions. The grid search method is used to optimize the penalty factor C and the radial basis function parameter γ. In two-dimensional space, the grid is divided for two parameter values, and each point of the grid is traversed to obtain the optimal parameter array. The specific steps are as follows:

The first step is to set the search range and the moving step size of parameters c and g. through research and comparison, the range of penalty factor C is set to $\left[2^{-3}, 2^{-1}, 2^{1}, 2^{3}, 2^{5}\right]$, and the search range of γ parameter is $\left[2^{-8}, 2^{-6} \cdots, 2^{6}, 2^{8}\right]$.

The second step is to test one by one according to the set moving step, and to test the performance of the classifier under each group of parameters by using the 10 fold cross validation method.

The specific steps of training classifier are as follows:

The first step is to input the positive and negative samples in the training set according to the saved path, respectively add category labels, and then input them into the preprocessing module, and the output is the training text vector represented by the feature weight.

In the second step, the generated text feature vector is brought into the classification function formula to solve its parameter value, and the specific form of the optimal classification function is obtained, that is, a two-class classifier is generated.

The third step is to set up the optimization range of penalty factor C and RBF kernel function γ value, establish the grid, calculate the classification function obtained by the corresponding parameter group of each point in the grid, and use the cross validation method to verify the classification effect.

4. Information classification method based on SVM organization modeling

After the introduction and analysis of SVM method in the previous chapter, combined with the characteristics of information text, we propose the information classification method and implementation steps based on SVM. The information classification based on support vector machine is to extract the content of information text, regard information recognition as two classification problems of text, use support vector machine method to train and learn the training information set, and get the classification model and decision function. Then the classification model is used to classify the classified information and identify the attributes of the information, that is, the process of judging whether the information belongs to garbage information. Its concrete realization is mainly divided into two stages, namely the training stage and the classification stage. As shown in Fig. 5:

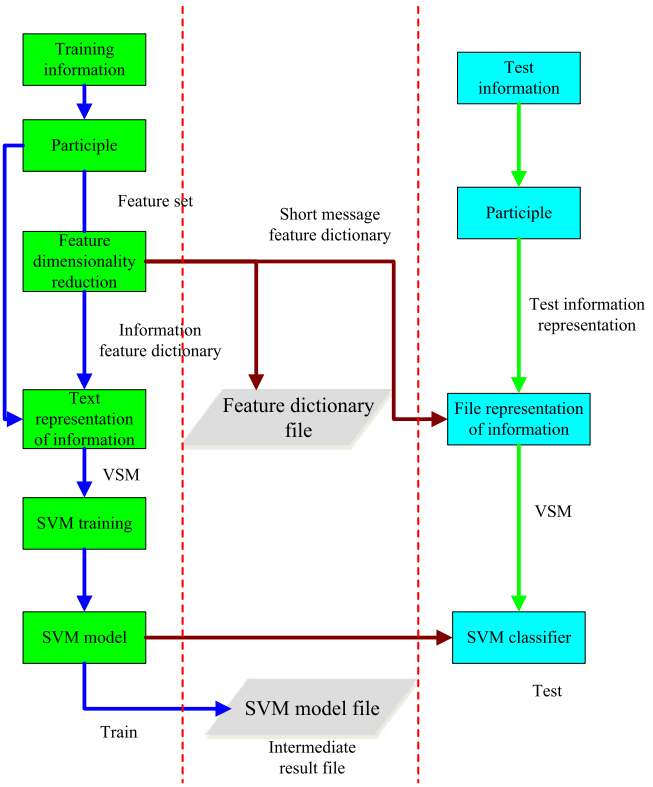


Fig. 5. Process chart of information classification based on support vector machine.

The information classification process based on SVM needs four steps: information preprocessing, feature dimensionality reduction, information text representation and classifier construction. First of all, the information training set and test set (i.e. known sample information and unknown classification information) should be preprocessed, the attribute categories and terms (i.e. features) of the information should be counted, and then the dimension of the feature set should be reduced to complete the vector representation of the features. In the process of training and learning the sample feature set, the parameters are calculated according to the sample points in the training set. In the next classification stage, unknown samples are classified by the decision function of the classification model to determine whether the information to be classified belongs to garbage information.

Information feature dimensionality reduction and classifier construction

After preprocessing, the information text often presents a high-dimensional feature distribution, and the feature space dimension composed of feature items is often very high. In this case, direct training and learning, on the one hand, the amount of calculation will increase with the increase of the feature dimension, thus "dimension disaster" will appear. Generally speaking, the main purpose of dimensionality reduction in feature space is to extract feature sets that contribute a lot to classification and improve the efficiency and accuracy of classification.

SVM is a supervised machine learning algorithm, which can be used in classification or regression problems. It uses a technology called kernel function to transform data, and then based on this transformation,

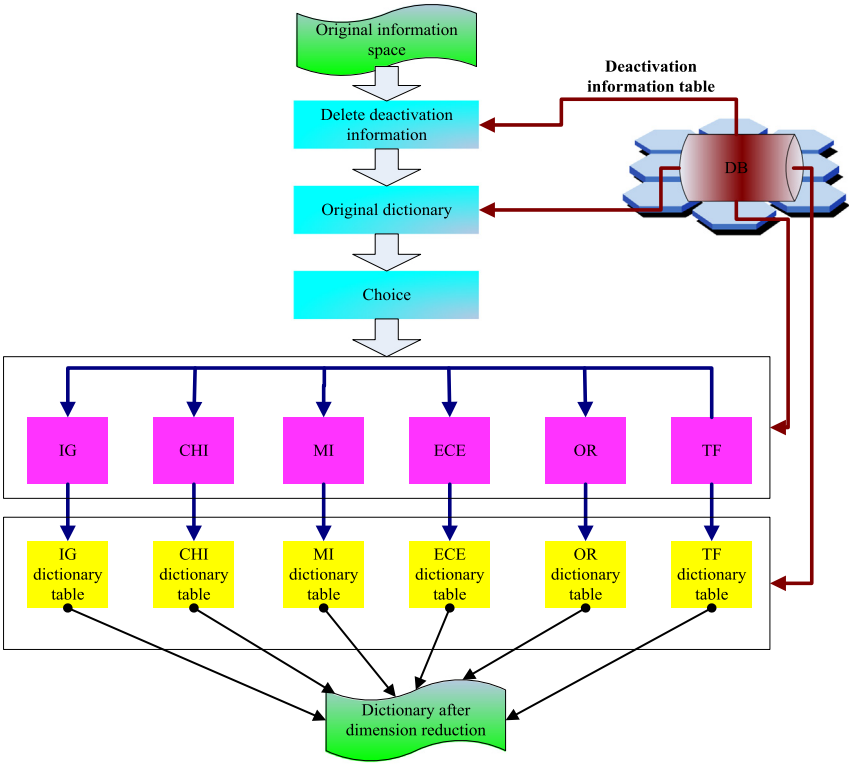


Fig. 6. Design of feature dimension reduction module.

the algorithm finds the best boundary between two possible classifications. In short, it makes some very complex data transformation, and then finds the method to distinguish data according to the defined label. As shown in Fig. 6, after the preprocessing process of information text, the original feature space formed has high dimension and sparsity of features, which requires calculating the information amount of terms for classification according to the feature dimensionality reduction method, and then sorting the terms to determine the new feature set.

The feature space obtained by different dimensionality reduction methods directly affects the classification results. However, the same feature dimensionality reduction method is affected by corpus, segmentation standard and classification algorithm, and the effect of classification is same. Therefore, the feature dimensionality reduction method in information text classification and recognition must be based on experiments. When the segmentation standard and classification algorithm are determined, the classification effect of different feature dimensionality reduction methods is compared to determine the best feature dimensionality reduction method. The specific steps are shown in Fig. 7:

Feedback learning of support vector machine

Because garbage information is not clearly defined, and its form and content are constantly changing, the garbage information filtering system needs to update the classification standards according to the new filtering requirements. With the dynamic change of garbage information characteristics and the need of adaptive filtering methods, feedback learning becomes an important part of filtering system. On the problem of feedback learning, the main methods are relearning and incremental learning.

Relearning is to form a new training set from the learned information and the information to be learned. At the same time, it initializes the classifier and relearns the training set on this basis. The specific implementation is to embed a trigger device in the information classification system. When the set number of information is received, the trigger device is activated and the learning is completed automatically. The advantage of relearning is that the algorithm is simple and easy

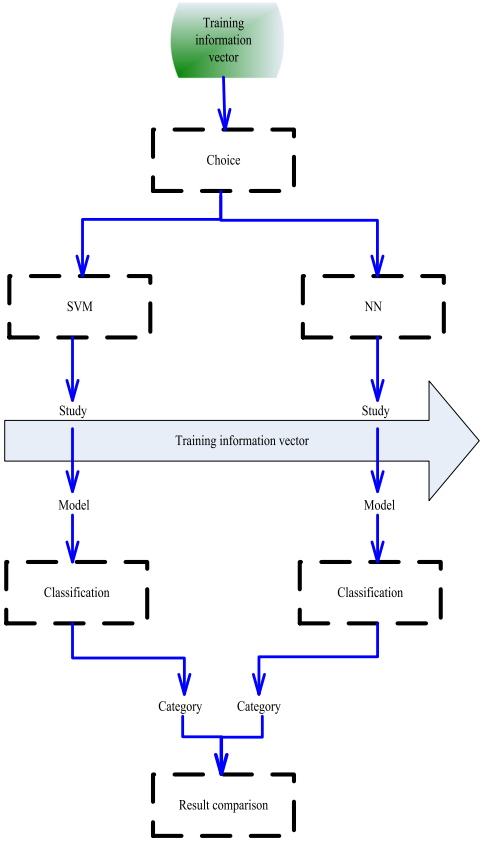


Fig. 7. Classification design process.

to implement, and it can adapt to the dynamic changes of information classification standards.

Incremental learning means that on the basis of the original classification learning, it does not need to scan the learned results again, and it can optimize itself by using the learned results or intermediate results. This learning method can be divided into one-time learning and batch learning. One study at a time means to consider one piece of information each time and complete the study several times. Batch learning refers to multi batch learning, considering multiple pieces of information at the same time, updating the classifier at one time.

The classification feature standard of information often changes with the change of demand, which needs to give up some features on the basis of the original standard features of information, and at the same time need to introduce new information features for feature reconstruction. The advantage of incremental support vector machine learning is to minimize the distance between features and attributes and the error penalty term by calculating the prior and the probability of attribute categories, and constantly modify the learned rules. Therefore, with the change of information classification standards, when some features are no longer the features of spam information, support vector machine method not only recognizes and classifies the information, but also calculates the probability of prior conditions and information attributes, adjusts the information value, and adds the learned features to the keyword list to complete the feature reconstruction. Compared with other methods, the learning process of SVM is simpler and the updating of features is more direct. Therefore, this paper uses incremental learning of support vector machine as the feedback learning method of garbage information filtering system.

The idea of information classification for SVM is to judge the feature items located at the information attribute boundary. The main task of classification is related to the support vector at the information attribute boundary, while other feature items do not play a role in classification. Therefore, for the feedback learning of SVM classification, the normal information after classification and recognition indicates that the model already has the classification information of the information, which has little value for feedback learning. However, the misclassified information has the classification information that the original classification model does not contain. The amount of these information may be very small, which is the focus of feedback learning. In the process of information classification, the actual situation is that the amount of normal information is often larger than garbage information, which also provides a practical advantage for feedback learning.

The information classification of SVM can be divided into training set, test set and to be tested set according to the different properties of information set. Different information sets have different feedback learning. According to the different relationship between information attribute and support vector, we call feedback learning of information training set as closed learning, and call information test set and prediction set as open feedback learning. The specific processing method is to conduct closed test on the information training and test set through the classification model, and manually mark the misjudgment information in the result. Information training set and test set feedback processing are mainly used to find the original misjudgment information, so as to correct the error information of training set, and then correct the noise of classification model. The information to be tested set refers to the unknown information set which needs to be classified in the actual information recognition. The feedback of the set to be tested is to store the information with wrong classification and high attribute value into the feedback set by considering the classification result obtained by the classifier. At the same time, the feedback set and the original classification model need to be optimized and trained to eliminate the influence of the feedback data on the classification performance.

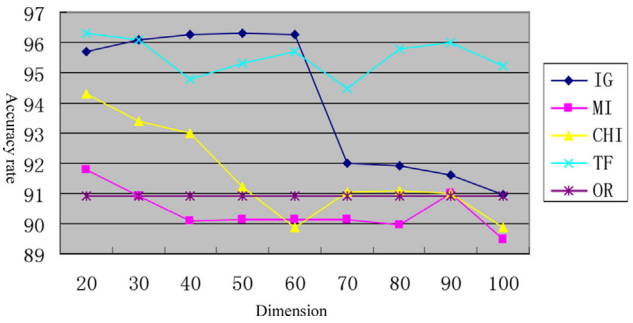


Fig. 8. Comparison of experimental results of feature dimensionality reduction methods.

5. Experiments and results

5.1. Experimental data and evaluation criteria

The experimental data of this paper is 2200 real SMS provided by a mobile communication operator, including 2000 normal SMS sent by users and 200 spam SMS confirmed by operators. The data source is reliable and true, which conforms to the experimental standard. The test set consists of messages with unacknowledged category attributes. The former, through training and learning, sums up the characteristics of SMS categories, and then constructs a classifier. The latter tests the test SMS through classifier to evaluate the effect of classifier. Usually in the experiment of text classification, the performance of the classifier is guided by the evaluation criteria. Accuracy and recall are used to measure the performance of the classifier. The accuracy rate is the proportion of all the correct text judged by the classifier to the actual correct text.

In the actual SMS filtering process, the principle should be to filter as many spam messages as possible without deleting the normal SMS. In other words, it is to reduce the risk of misjudgment and improve the accuracy of SMS classification as much as possible. Therefore, in this experiment, the accuracy is used to measure the performance of the classifier. The evaluation standard formula of this experiment is as follows:

$$\text{Accuracy rate} = \frac{\text{Number of messages classified correctly}}{\text{Total SMS}} \times 100\% \qquad (13)$$

B. Experimental results and analysis of feature dimension reduction method

In the process of SMS classification, the selection of dimensionality reduction method has a huge impact on the classification effect. We choose five feature selection methods: mutual information (MI), feature frequency (TF), probability ratio (OR), information gain (IG), x2 Statistics (CHI) to carry out the comparative experiment of short message classification. The common feature of these feature reduction methods is to extract important key features and form a new low dimension space without changing the original feature space, so as to reduce the difficulty of classification. Their experimental results are compared as follows:

As shown in Fig. 8, the effect of information gain and feature frequency method is better than the other three methods, followed by x2 statistical method, and the classification efficiency of mutual information method is the worst. The accuracy of information gain method decreases with the increase of feature dimension. Through comparative analysis, we can see that the information gain method is the best when the feature number is 50 dimensions. After 50 dimensions, with the increase of feature number, the accuracy of classification decreases rapidly. The reason may be that the information gain method takes into account the situation that the entry does not occur, which makes the classification efficiency of the information gain method unstable when taking different feature dimensions.

Table 1
Experimental results of kernel function.

	Information set	Normal information	Spam information	Whole
Linear support vector machine	Training set	94.7%	74%	83.9%
	Test set	96.6%	70%	82.76%
Polynomial kernel support vector machine	Training set	92%	82%	86.46%
	Test set	93.8%	54%	73.46%
Sigmoid kernel support vector machine	Training set	87.8%	62%	74.36%
	Test set	92.3%	63%	77.22%
Radial basis kernel support vector machine	Training set	98%	93%	94.96%
	Test set	96.2%	91%	93.16%

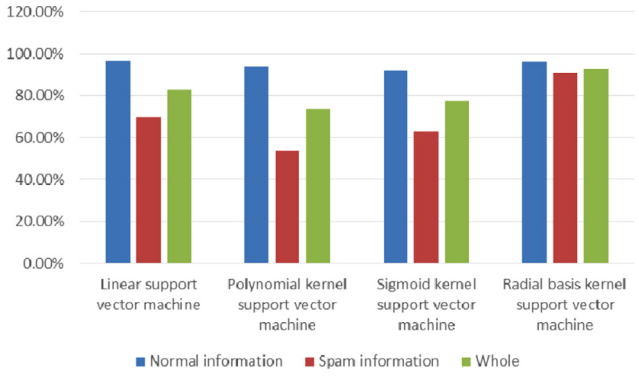


Fig. 9. Test results of different methods.

Through the experimental comparison of feature dimensionality reduction methods, we get that the information gain method is relatively efficient in short message classification. Therefore, we choose the information gain method as the feature dimensionality reduction method of short message classification in this paper. At the same time, the first 50 dimensions are selected as the feature number of information gain.

Experimental results and analysis of kernel function

In general, in the process of SVM classification, it is necessary to map samples to high-dimensional feature space through nonlinear mapping, so that the problem can be divided linearly or approximately linearly, which requires the introduction of kernel function to avoid directly solving unknown functions. We do some experiments on the above four kernel functions. The experimental results of kernel function are shown in Table 1.

The test results of different methods are compared as shown in Fig. 9. We can see from the experiment that the radial basis kernel SVM is the most accurate method for short message classification. The experiment also shows that the recognition accuracy of normal SMS is higher than that of spam SMS. The principle of SMS filtering is to filter junk SMS without affecting normal SMS as much as possible. The classification accuracy of SVM meets this requirement.

Experimental data comes from UCI self machine learning database, which are collected by relevant experts in the process of clinical treatment. Data set (1) is coronary heart disease data set. There are 573 samples in the whole dataset, including 76 attributes and 4 classifications. After preprocessing, 13 attributes and 2 categories of complete samples were obtained. There are 339 training samples and 234 test samples. Data set (2) is lung cancer data set. There are 683 samples in the whole data set, including 11 attributes and 2 classifications. After preprocessing, 10 attributes and 2 classifications of complete samples were obtained. There are 390 training samples and 293 test samples. Through the experiment, it is found that the classifier is very effective for the classification of medical related data sets, and can play a certain auxiliary role in medical diagnosis. At the same time, it is found that data preprocessing, especially data normalization, is the key step to improve the classification accuracy.

5.2. Experimental comparison between support vector machine and neural network

We compare the two methods of SVM and neural network, and the results are as follows Table 2 and Fig. 10.

The experiment shows that the accuracy of SVM is 94%, which is better than neural network. Support vector machine method is based on the principle of structural risk minimization, while neural network method is based on the principle of empirical risk minimization. We

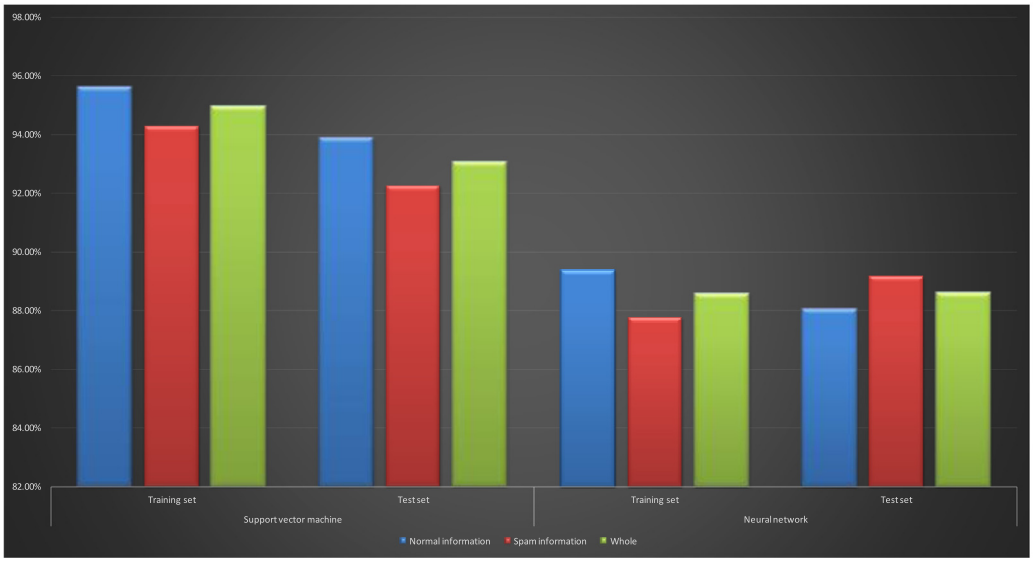


Fig. 10. Test and training comparison results.

Table 2
Experimental results of support vector machine and neural network.

	Information set	Normal information	Spam information	Whole
Support vector machine	Training set	95.63%	94.29%	94.96%
	Test set	93.88%	92.22%	93.06%
Neural network	Training set	89.38%	87.76%	88.57%
	Test set	88.05%	89.15%	88.61%

think that the difference of training process is the basic reason why SVM is better than neural network.

6. Conclusion

The information classification method based on SVM is proposed, and the key technologies designed are studied and analyzed, including feature dimension reduction method and information classification algorithm, and the penalty parameters and kernel functions suitable for information classification of support vector machine are selected through experiments. In this paper, we do some research on information filtering mining methods, but we also find some shortcomings and problems in the process of research. In this paper, the parameters and feature dimensionality reduction methods of SVM are selected through experiments. How to select the optimal parameters and feature dimensionality reduction methods of SVM based on theory is the next problem to be studied. The information filtering method of the Internet of things big data based on SVM proposed in this paper focuses on the recognition and filtering of garbage information, and does not consider the impact of information recognition processing speed on the filtering effect. The next step is to introduce the processing speed of information recognition into the research of garbage information filtering and try to use the core technology of cloud computing, Hadoop platform. It takes HDFS as the system management storage, and MapReduce as the model for cloud computing operations. Based on this, it constructs a platform for massive data mining, analysis and sharing. It has become a new trend of data mining technology in recent years to build a system of massive data mining algorithm with Hadoop as the core.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by Zhejiang Basic Public Welfare Research Project(Research and Implementation of Personalized Service System for Home-based Old-age Care Based on Data Fusion: LGF18F020004).

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